

Register · The ARC Theory · The Dated Prediction Register: Forward Predictions as One Verifiable Chain

Every dated prediction of the programme in one chain, from a sealed manuscript bundle of 8 December 2024 to the staged draft registrations, with each prediction's date, the kind of record that dates it and its status.

Michael Darius Eastwood · Working Paper v2.1 · first published 25 August 2026 · revised 5 October 2026 · No part of this paper is peer reviewed; it is a dated preprint. Companion page: the dated predictions register (<https://www.michaeldariuseastwood.com/research/dated-predictions/>). Machine twin: design-embedded-predictions-register-`.json` (<https://www.michaeldariuseastwood.com/research/data/design-embedded-predictions-register.json>).

Abstract. Five independent cold-model assessments of this programme converged on one criticism: that its predictions were not registered in advance, and that its confirmations might therefore be accommodations. This working paper assembles the programme's forward predictions as a single dated chain and states exactly what that chain does and does not establish. It does not make the programme preregistered in the registry sense, and it never uses that word on its own for anything except an accepted registry submission: a design fixed by a dated timestamp before its run is called preregistered by timestamp. It establishes something narrower and checkable: every core quantitative claim has at least one dated artefact fixing it before the measurement that tested it, with the exception of Paper VII (<https://doi.org/10.17605/OSF.IO/X6WA7>)'s fifty-domain cohort, whose predictions were not timestamped in advance of their own fits, and a stranger can verify each link in minutes from public materials. The register's claim is deliberately an ensemble claim: any single prediction here may have been stated earlier by someone else, and where that is known it is listed as convergence rather than threat; what no earlier record contains is all of these statements together, in one record, before the programme's experiments ran. The chain runs from a sealed manuscript bundle of 8 December 2024 through a printed edition (<https://www.michaeldariuseastwood.com/book/>) of 2 January 2026 carrying nine numbered predictions with falsification criteria, a public in-paper priority ledger of 13 February 2026 (<https://doi.org/10.17605/OSF.IO/HQCGF>) with named falsifiers, a live laboratory notebook (<https://www.michaeldariuseastwood.com/research/reports/arc-align-scaling-report.html>) commenced 10 March 2026, result files of 11 and 12 March 2026 whose headers freeze designs and planted answers before responses, a tiered 50-domain suite of 16 March 2026 with per-domain predicted families, and a complete preregistration folder of 17 March 2026 whose locked manifest fixed twelve predictions before their fits, to the programme's staged draft registrations and its standing unproven wagers. The chain's sharpest artefact is a correction: public commit history shows the author demoting his own successful 10/12 result from “pre-re-

gistered” to “pilot dry run” thirteen minutes after recording it, months before any external critic existed. Version 2.0 adds a census of every dated prediction in the programme (Section 8): the theory registration’s twenty-two propositions with their registers and kill conditions, the registered corrector-class audit, each draft design preregistered by timestamp, the printed predictions and the predictions fixed inside designs, each with its date, the kind of record that dates it and its status.

1. The doctrine, stated exactly

Three vocabulary rules govern every sentence of this paper. First, as amended on 3 October 2026, *registered* is kept for a registry submission that has been accepted, and a design fixed by a dated timestamp before its run is *preregistered by timestamp*, with its date and receipt, which has been the programme’s usage since 24 September 2026; as read on 4 October 2026, one of the programme’s studies has an accepted registry submission, the submission click is a human act, and no tool or agent performs it. Second, the printed book (<https://www.michaeldariuseastwood.com/book/>) is a *prior statement*, never a preregistration: print fixes a prediction, but only a registration fixes an analysis plan. Third, forward statements and retrospective matches are never summed: this register contains only dated forward statements, and retrospective convergences live separately in the convergence register (<https://www.michaeldariuseastwood.com/research/convergences/>), typed by evidence class. The state under the first rule, as read on 4 October 2026, is this. Two registrations have been accepted. One is the theory registration (<https://doi.org/10.17605/OSF.IO/P8CKQ>) of 8 September 2026, which registers the theory’s twenty-two propositions and is not the registration of any study. The other is the registration of one of the trials, the corrector-class audit (unit o6), registered on OSF on 1 October 2026 and public since 4 October 2026, at the end of OSF’s approval period (DOI 10.17605/OSF.IO/RV2B5 (<https://doi.org/10.17605/OSF.IO/RV2B5>)). Every other trial is a draft awaiting the author’s submission. Sixty-five of the programme’s designs, unit o6 among them, are preregistered by timestamp, the first of them anchored on 12 August 2026 at Bitcoin block 962066.

The claim made here is the conjunction. Each part may be conceded individually without cost: someone may have predicted any one of these things earlier, a folder is not a registry, a book is not an analysis plan. The dated co-occurrence of all of them in one record before the programme’s experiments ran is the claim, and it is refutable: produce an earlier single record containing this ensemble, or show any link’s date to be wrong.

2. The chain

DATE	ARTEFACT	WHAT IT FIXES	STATUS
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8 Dec 2024	Sealed manuscript bundle, five attachments, 221,236 words; self-emailed, date per the sender-copy Date header; SHA-256 authenticates the bytes; received copies carry the authenticated header chain	The framework's core commitments in writing before any experiment	dated private record
30 Apr 2025	Intermediate manuscript	Earliest verbatim anchor for several later-published passages; corroborated on live pages	dated private record
2 Jan 2026	Print edition, ISBN 978-1806056200 (https://www.michaeldariuseastwood.com/book/): Appendix F (five numbered predictions) and Appendix A (§A.3 four further predictions; §A.4 falsification criteria)	Nine numbered predictions with deadlines, thresholds or falsifiers, in print	printed public record
13 Feb 2026	Paper III (https://doi.org/10.17605/OSF.IO/HQCGF) 's Priority Record section, publicly dated, falsifiers F4, F7, F10 to F12	Named predictions including the composition-operator forward test and the leaf venation $d/(d+1)$ exponent (whose 10 March correction is itself dated)	public dated ledger
10 Mar 2026	The live laboratory notebook (https://www.michaeldariuseastwood.com/research/reports/arc-align-scaling-report.html): “Date Commenced: 10 March 2026 · Document Version: Live - updated in real time”	The experiments recorded as they ran, failures included (v4 false positives exposed at v5 blinding)	live dated record

11-12 Mar 2026	Thirty-six result files in the public repository (https://github.com/MichaelDariusEastwood/arc-principle-validation)	Depth configurations, planted expected answers, four-layer blinding protocol and scorer counts frozen in file headers, timestamped to the second, before the responses they score	design-embedded
16 Mar 2026	Tiered 50-domain suite result files	Per-domain predicted families; empirical tier 19/25 original candidate set, 18/25 corrected seven-model rerun of 11 August 2026, both reported, tiers never blended	exploratory, both counts reported
17 Mar 2026	Complete preregistration folder (protocol, locked 12-row manifest, checksums, registration text) and the commit sequence e5c22dd (https://github.com/MichaelDariusEastwood/arc-principle-validation/commit/e5c22dd) 00:43, f5e06a8 (https://github.com/MichaelDariusEastwood/arc-principle-validation/commit/f5e06a8) 00:47, c853277 (https://github.com/MichaelDariusEastwood/arc-principle-validation/commit/c853277) 00:56	Twelve predictions fixed before their fits; then the author's own thirteen-minute demotion of the 10/12 result's status	demoted by the author; reversed in one half only on 7 Sep 2026, the twelve predictions having been timestamped in advance of their outcome (section 4)
12 + 23 Aug 2026	OpenTimestamps vintages; public proofs in research/data/timestamps/ (https://www.michaeldariuseastwood.com/research/data/timestamps/)	Bitcoin-attested anchors of the hash manifests; an anchor dates bytes at its own vintage, never earlier	cryptographic anchor

25 Aug 2026	Eight complete draft registrations staged; 80-domain expansion registration prepared, catalogue frozen by SHA-256	The analysis plans; no fit has ever run on the eighty	as read on 4 Oct 2026: of the eight, the theory registration (https://doi.org/10.17605/OSF.IO/P8CKQ) was accepted on 8 Sep 2026; the corrector-class audit (unit 06) was registered on OSF on 1 Oct 2026 and has been public since 4 Oct 2026 (DOI 10.17605/OSF.IO/RV2B5 (https://doi.org/10.17605/OSF.IO/RV2B5)); the other six await the author's submission
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3. The printed appendices, graded severally

Appendix F closes: “These predictions are my wager. If they fail, the framework is wrong or incomplete. If they succeed, something important has been glimpsed. Time will judge.” The programme grades the five predictions unequally, because claiming them equally would be false precision. Prediction 2 (alignment drift above 15 per cent within 18 months without hardware-level constraint, below 5 per cent with it, difference significant and replicable) is preregistration-grade in substance: quantitative thresholds for both arms, a time window, an inferential commitment. The prepared study-v draft registration carries those printed thresholds unchanged into its formal hypothesis, citing the appendix by name: the book made the prediction, the registration supplies the measurement, and the numeric commitment predates the study that will test it. The theory registration (<https://doi.org/10.17605/OSF.IO/P8CKQ>) states the limit of that draft: Prediction 2 is “the subject of its own drafted unit, with the hardware claim and the eighteen-month window declared untested inside it”. Predictions 3 and 4 are near-grade (a threshold and deadline without named benchmarks; a comparison and method without a threshold). Predictions 1 and 5 are predictions only, and Prediction 5 is graded the weakest, vague enough to be claimed either way, so it is held to the strictest standard before it may ever be claimed.

Appendix A is graded separately and less kindly, because its predictions and its reasoning do not stand together. Section A.3 carries four predictions, each with a variable to vary and an outcome to measure, and the fourth is a controlled design in the ordinary sense: identical training data, different sequencing of value-relevant examples, persistence of the early values measured against the later ones. Section A.4 states four falsifiers in advance. But section A.2, which is where the square is justified, argues from three phenomena it calls exponential to a relation that is quadratic, and its third argument makes the slip explicit by describing compounding at one per cent a cycle as “the exponential behaviour predicted by R^2 ”. Exponential growth and quadratic growth are different classes and the quadratic is the slower of the two, so the section’s supports do not reach its conclusion. The prediction is dated, public and genuine; the printed derivation beside it is not sound, and the programme’s own chronology says as much in saying the book printed the square without deriving it and dating the formal open exponent to 17 January

2026. The measurement has not supported the square: on Gemini 3 Flash, Paper II (<https://doi.org/10.17605/OSF.IO/8FJMA>) reports an endpoint sequential exponent of 0.59 with a bootstrap 95 per cent interval of minus 1.3 to 2.9, a regression sequential exponent of 0.49 with a standard error of 0.20 and a coefficient of determination of 0.86, and a compute-matched parallel exponent of 0.31. The point estimates are sub-linear, the fit is one model's and inconclusive, and the 2.24 that would have supported a square was publicly retracted, which is why the outcomes register grades the quadratic prediction open-undecided rather than fulfilled. Appendix A's third prediction and third falsifier rest on Google Willow, which carries both dates wherever it appears here: it was announced with its paper in Nature (<https://doi.org/10.1038/s41586-024-08449-y>) on 9 December 2024, and the team's preprint (<https://arxiv.org/abs/2408.13687>) was public from 24 August 2024, before the December manuscript record, so it is graded convergent timing and never prediction.

4. The thirteen minutes

On 17 March 2026 at 00:43, commit e5c22dd recorded "Pre-registered 12-domain extension: 10/12 confirmed ($p = 5.44e-4$)". At 00:47, f5e06a8 adopted a conservative miss posture. At 00:56, c853277 downgraded the run to a pilot dry run. The result was never retracted; ten of twelve stands in the result files today. What was corrected, by the author, unprompted, in public, months before any external critic existed, was the claim about the result's status. The packet even instructs against its own over-claim: "Do not upload this packet unchanged as a preregistration. Use it only as an audit trail for the first locked 12-domain dry run." The demotion is kept here exactly as it was written, and it is reversed in one half only: the twelve predictions were committed to the public repository at 00:19:19 UTC on 17 March 2026, attestation class public repository commit date, twenty-three minutes before the fits ran, so they were timestamped in advance of their outcome. The reversal of 7 September 2026 runs under the attestation-class rule, which does not demote a prediction for the absence of a registry form; what the reversal does not reach, and what still stands, is the packet's own written condition that it be deposited to a registry before data extraction, which was not met. One row repeats a domain, operator class and predicted model already public at 23:54:53 UTC on 16 March 2026, attestation class public repository commit date, with a matching outcome, so the strictly fresh count reads 9 of 11, $p = 1.372e-3$. This is the programme's correction discipline operating at minute resolution, and it is why the honesty record and the prediction record are the same record.

5. What this register does not establish

It does not make the programme preregistered in the registry sense; no completed study has an accepted registration, and this paper never claims otherwise. The March 2026 folder governs only the 12-domain extension; its own registration text states it does not retroactively preregister the 25-domain or 50-domain cohorts, and that criticism stands. It does not establish priority over other authors for any individual idea; earlier statements by others are convergence and are listed as such. It does not establish that the predictions are true: the standing wagers below are open, and several carry deadlines that can simply fail.

6. Standing wagers, open and dated

PREDICTION	DATED	STATUS
Alignment drift 15/5 per cent within 18 months (Appendix F, prediction 2)	2 Jan 2026	Untested; study-v prepared
Recursive capability gains above 300 per cent by 2029 (Appendix F, prediction 3)	2 Jan 2026	Open
Meta-cognitive emergence by 2028 (Appendix F, prediction 1)	2 Jan 2026	Open
Value stability under adversarial conditions (Appendix F, prediction 4)	2 Jan 2026	Untested
Convergent consciousness signatures (Appendix F, prediction 5)	2 Jan 2026	Open; held to the strictest standard
80-domain catalogue: frozen family predictions beat the prevalence baseline by 0.20 under both classifications	catalogue Mar 2026; registration prepared 25 Aug 2026	Never fitted
Fourth-cell (logarithmic) test	prepared Aug 2026	Awaiting human submission
$\gamma = 1/2$ (conversion exponent)	dated papers	Conjecture, stated as such

7. Verification

Every chain link names its artefact. The repository is public and its commit history cannot be backdated without detection; the print edition (<https://www.michaeldariuseastwood.com/book/>) is fixed by ISBN; the OpenTimestamps proofs are Bitcoin-attested; the machine-readable register carries all thirty-six run-file rows with per-artefact paths and date bases. A reader who trusts none of the prose can start from the JSON (<https://www.michaeldariuseastwood.com/research/data/design-embedded-predictions-register.json>), open the named files, and check the dates.

8. The census: every dated prediction in the programme

Sections 2 to 6 follow the chain by date. This section gathers in one place every prediction the programme has dated, so that nothing a reader looks for is missing from the register: the theory's twenty-two registered propositions, the registered unit, the draft designs preregistered by timestamp, the printed predictions and the predictions fixed inside designs. Each entry carries its date, the kind of record that dates it and its status as the record states it today. Listing a prediction here confirms nothing; a status changes only when a registered test reports.

8.1 The theory's twenty-two propositions

Registered on OSF on 8 September 2026 at version 1.100 and updated at version 1.102 on 13 September 2026 (the theory registration (<https://doi.org/10.17605/OSF.IO/P8CKQ>), DOI 10.17605/OSF.IO/P8CKQ); both texts are frozen. Each statement and each kill condition below

is quoted from the registration. It assigns fifteen of them to five registers, ARC-1 to ARC-5, each of which tests one of the theory's five laws. Laws I to III were registered on 8 September 2026. Laws IV and V, the ARC Persistence Law and the ARC Embedding Law, have stood in the theory since 24 September 2026: their three propositions are registered, but neither law is registered as a law and neither has been tested, and their mathematics is set out in Laws IV and V: the mathematics, version 1.0 (<https://www.michaeldariuseastwood.com/research/reports/laws-iv-v-mathematics-v1-0.html>) (3 October 2026). The other seven belong to no register. All twenty-two stand at the first rung of the evidence ladder: registered, not yet tested.

PROPOSITION	REGISTER AND LAW	REGISTERED STATEMENT	KILL CONDITION	STATUS
P1 Power-law conversion and recursive-depth advantage	ARC-1, Law I (the ARC Principle)	“Within the scope, usable capability is best described by a power law in modelled recursive depth, and the depth exponent exceeds the compute-matched and token-matched breadth exponent by the registered margin. Whether it also exceeds one is P8's separate verdict.”	“a registered fit in which a rival form beats the power law under the registered model comparison across a majority of estimable cells.”	registered, not tested
P2 The sustainable growth profile has an interior maximum	ARC-1, Law I (the ARC Principle)	“Within the scope, the sustained growth exponent does not rise without limit as reinvestment rises, and the profile has a maximum inside the administered range. This is a claim about the shape of that profile and is firewalled from the framework's boundary claim, which P16, P20 and P3 carry.”	“a profile that rises throughout the administered range without turning over, where the registered shape adjudication returns monotone rather than unimodal. The adjudication rule is the titration design's registered rule and is restated here so it cannot drift: an interior maximum is declared only when the bootstrap interval on the location of the maximum excludes BOTH ADMINISTERED ENDPOINTS, the lowest as well as the highest, and the registered unimodality check passes; otherwise the profile is adjudicated monotone in whichever direction the registered shape rule returns, and a maximum located at either boundary refutes.”	registered, not tested

P3 The corrected relation upper-bounds the sustainable frontier	ARC-3, Law III (the ARC Ceiling)	“The maximum growth exponent that a system holds while remaining correctable lies at or below the value the ceiling relation returns from the correction-leverage exponent measured on that same system, and where that exponent is one half the value is two.”	“an interval on the profile maximum lying entirely above the derived value, which replicates on fresh data at the same settings, and which is obtained while the system is still correctable, meaning its correction exponent out-scales its drift exponent with the lower bound of that margin above zero across a window fixed before the run.”	registered, not tested
P4 Correction out-scaling drift is what carries the asymptotic advantage	ARC-2, Law II (the ARC Co-Scaling Law)	“Within the registered minimal model, relative correctable burden tends to zero where the correction exponent exceeds the drift exponent, burden holds the asymptotic advantage where the correction exponent is the smaller, and at equality the outcome, in the finite run and in the long run, is decided by coefficients, delay, backlog and saturation rather than by the exponents.”	“an observed system that remains aligned on the registered measure while its measured correction exponent sits below its measured drift exponent, over a horizon long enough for the shortfall to have shown.”	registered, not tested
P5 The exponent is derived, not fitted	no register: the derivation’s status	“The growth exponent implied by an independently measured coupling agrees, within a registered tolerance, with the exponent fitted from the growth curve.”	“systematic disagreement beyond tolerance across a majority of estimable systems.”	registered, not tested
P6 The correction exponent is bounded for same-class correctors	ARC-3, Law III (the ARC Ceiling)	“Where a system corrects itself using mechanisms of its own class, the correction exponent is at or below one half.”	“a same-class corrector whose measured correction exponent interval lies entirely above one half.”	registered, not tested

P7 Deployed correction is of the weaker class	no register: a survey of current practice	“The correction mechanisms actually used in deployed alignment practice predominantly belong to the class whose strength is fixed by construction rather than the class that can scale with the capability it corrects.”	“a preregistered classification in which the scaling class predominates.”	registered, not tested
P8 Measured exponents exceed the null	ARC-1, Law I (the ARC Principle)	“Across the admissible independent systems the pooled growth exponent exceeds the one-half null; an interval lying wholly below the LOWER EDGE of the registered equivalence region refutes rather than supports it, where an earlier wording said wholly below one half and so contradicted the four-outcome rule in P8's own entry, and whether it also exceeds one is the separate CLEARS UNITY verdict.”	“a pooled interval on the growth exponent lying wholly below the LOWER EDGE of the registered equivalence region around one half.”	registered, not tested
P9 The cross-domain form	ARC-1, Law I (the ARC Principle)	“Where an effective dimension can be assigned to a domain, the growth exponent follows the dimensional relation with the dimension supplied by that domain's equation of state.”	“measured exponents disagreeing with the dimensional prediction across two or more independent domains.”	registered, not tested

P10 Scorer-family dependence	no register: the scoring instrument	“A material fraction of the results this programme has scored unblinded within a single model family, ten per cent, fixed in the entry below, fails to survive re-scoring by a different family.”	“a survival proportion whose interval excludes a material loss, where material is fixed now at ten per cent, which is a declared convention fixed before any result rather than a quantity this framework derives, so a fraction landing just either side of it is reported as a result at the margin: P10 is REFUTED when the 95 per cent interval on the fraction of earlier results that do not survive cross-family re-scoring lies entirely below ten per cent, SUPPORTED when it lies entirely above ten per cent, and INCONCLUSIVE otherwise. The re-scoring study's own registration carries the same threshold. A refutation here would mean this programme's blinding result does not reach its own earlier work, which is the one direction in which that result is convenient for its author.”	registered, not tested
P11 The residual-decay and correction-capacity exponents are not practically interchangeable	no register: the derivation's status	“the residual-decay and correction-capacity exponents are not practically interchangeable, and the working practice of treating them as one is an assumption rather than a fact.”	“measurements of the first two, taken on instruments that share no fitted quantity, agreeing within a registered tolerance across model families, which would license the identification the programme currently only assumes.”	registered, not tested
P12 Build order moves the correction-leverage exponent	ARC-5, the genesis strategy (Law V, the ARC Embedding Law)	“The order in which a system's components are built moves the correction-leverage exponent by at least the registered minimum effect.”	“a preregistered manipulation of build order producing no change in the correction-leverage exponent beyond the tolerance registered with that instrument, where the instrument has shown from its own sensitivity that a change of that size would have been detected. A manipulation too weak to move anything is not a refutation of this prediction; it is a failed instrument, and condition D above says so in advance.”	registered, not tested

P13 Same-family automated correction underperforms cross-family at matched resources, and through the measured error correlation	no register: corrector composition	“Where an automated alignment researcher post-trains a target model against alignment benchmarks, a researcher from a different model family than the target reduces held-out misalignment more than a researcher from the same family, at matched data and compute, and the panels differ in measured pairwise error correlation in the direction that would explain it. The two halves are one conjunctive proposition, and an advantage arriving without the correlation difference is reported as unexplained by this framework rather than as support for it.”	“the interval on the cross-family minus same-family difference in held-out misalignment reduction sitting wholly at or below zero, with data and compute matched and with scoring blind to condition.”	registered, not tested
P14 Gains shrink under blinded cross-family scoring	no register: the scoring instrument	“Where an automated researcher’s method is re-scored by a scorer that cannot see the condition and comes from a different model family than either the researcher or its target, the alignment gain it returns is smaller than the gain the unblinded same-family scoring reported.”	“the interval on the blinded minus unblinded gain sitting wholly at or above zero across the same items.”	registered, not tested
P15 Externally installed gains decay under subsequent capability training	ARC-4, value persistence (Law IV, the ARC Persistence Law)	“Gains installed by external post-training erode under later capability-directed training that embeds no correction of its own, and more than half of what was installed is lost once the capability-training budget matches the alignment budget that installed it.”	“an interval on the retained fraction of the installed gain, after matched capability training without embedded correction, lying entirely above one half.”	registered, not tested

P16 The prohibition itself	ARC-3, Law III (the ARC Ceiling)	“A system does not hold both at once: a growth exponent above the point at which its own measured correction margin reaches zero, and a correction margin whose lower bound stays above zero, sustained across a window fixed before the run.”	“a run holding the exponent above that point with the lower bound of the correction margin above zero across the registered window, replicated on fresh data at the same settings.”	registered, not tested
P17 Panel corrector failures are conditionally independent on a frozen fault set	ARC-3, Law III (the ARC Ceiling)	“Within a corrector panel named in advance by model lineage, substrate, training provenance and scaling class, pairwise joint misses on a frozen fault set of stated type are practically equivalent to the conditional-independence baseline under the registered margin; failure-mode overlap is the outcome rather than the entry criterion, and this is the cross-sectional claim the instrument can decide rather than the temporal premise it bears on.”	“a registered panel drawn from the population above, whose joint failure rate exceeds the independence baseline by the equivalence margin registered with that instrument, which that unit fixes at 0.20 on the log scale of the ratio of observed to predicted joint misses, an inflation of about 22 per cent over independence, on a frozen fault set whose type is stated. The panel and the fault set carry their type, checkable or judged, because P22 predicts the answer differs between them and a panel reported without its type cannot be read against either.”	registered, not tested
P18 The composition operator predicts the family	ARC-1, Law I (the ARC Principle)	“Classified in advance by how a domain composes its inputs and outputs, the scaling family that best fits that domain's data is the family the classification predicts, and not one chosen after the curve is seen.”	“the predicted family failing to beat, by the margin registered with the deciding instrument and across the registered domain set, the comparator that belongs to the draw actually used: the balanced rate where the domains are drawn balanced across families, and the empirical prevalence where they are drawn unstratified.”	registered, not tested

P19 External alignment does not scale with capability	no register: a survey of current practice	“Alignment achieved by mechanisms outside a system's own recursion does not improve as the capability of the system it governs rises: its exponent is indistinguishable from zero.”	“on the zero-scaling axis, which is this proposition as worded, an alignment exponent whose interval lies entirely outside the registered equivalence region of plus or minus 0.10 about zero, in either direction; and on the keep-pace axis, reported beside it, an interval lying entirely above one half.”	registered, not tested
P20 The ceiling relation takes the corrected form	ARC-3, Law III (the ARC Ceiling)	“Where a stability boundary can be located and a correction exponent measured on the same system, the boundary follows one over one minus the correction exponent rather than one over the correction exponent or a fitted constant.”	“the discrimination registered with the deciding instrument selecting a rival over the registered relation across a majority of admissible cells, on systems held out of the fitting.”	registered, not tested
P21 In-loop correction pulls away with depth	ARC-5, the genesis strategy (Law V, the ARC Embedding Law)	“Correction that participates in each subsequent revision round outperforms correction that sees only the finished output, and the advantage widens as recursive depth rises rather than staying constant.”	“no interaction between correction placement and depth beyond the tolerance registered with that instrument, in a design whose own sensitivity shows an interaction of that size would have been detected. A design too small to find the interaction is a failed instrument and not a refutation, as condition D already provides.”	registered, not tested
P22 Checkable correction carries the higher exponent, so the ceiling is lowest where the safety claim lives	ARC-3, Law III (the ARC Ceiling)	“The checkable correction-leverage exponent exceeds the judged correction-leverage exponent on the same systems under the same instrument, so the ceiling is strictly lower for judged correction.”	“a measured judged correction-leverage exponent exceeding the checkable one on the same systems under the same instrument, with the reversal replicating on fresh data at the same settings.”	registered, not tested

8.2 The registered unit

One study unit is registered: the corrector-class audit (unit o6), registered on OSF on 1 October 2026 and public since 4 October 2026, at the end of OSF's approval period (DOI 10.17605/OSF.IO/RV2B5 (<https://doi.org/10.17605/OSF.IO/RV2B5>)). In its own words it is “the instrument the theory

registration (version 1.102, DOI 10.17605/OSF.IO/P8CKQ) names as deciding P7”, a survey of current practice that the registration places in no register and names as the premise of the Eden Protocol. Its three registered predictions:

- **The class distribution.** “The prediction is that at least 90 per cent of the classified, non-indeterminate deployed subset falls in class 1, static, or class 2, capability-carried, and that class 3 mechanisms are absent from deployed production systems or confined to research prototypes.”
- **Decidability from public documentation.** “The prediction is that at least 80 per cent of inventoried mechanisms receive an agreed class when two blind assignments are made, the second without sight of the first.”
- **Exhaustiveness of the taxonomy.** “The prediction is that no inventoried mechanism requires a fifth dependency.”

Status: registered, not yet run. Its census window closes at 23:59 UTC on 11 October 2026, as registered, and its run is the author’s.

8.3 The draft designs, preregistered by timestamp

The other study designs are draft registrations awaiting the author’s submission, apart from those held from publication and the one withdrawn before filing. Most are preregistered by timestamp: OpenTimestamps receipts in Bitcoin anchor their files, the first on 12 August 2026 at block 962066, and a design’s own receipt, where it has one, dates it. As the registration diligence table (<https://www.michaeldariuseastwood.com/research/data/registration-diligence-table.json>) records them, there are 67 drafts by unit and stage below, with 2 further designs held from publication and 1 withdrawn before filing. None of these drafts is registered on OSF, and none has run. Each unit keeps the number and name it was first given.

UNIT	DESIGN	STAGE
01	study-eclipse: gamma ratio direct	draft (working)
02	study-k: arc bound	draft (working)
03	study-ai: boundary mapping alpha gamma frontier	draft (working)
05	study-ad: correction exponent estimation	draft (working)
08	study-ah: correlated failure same substrate	draft (working)
09	paper-x: coupled coscaling correction	draft (final and ready)
10	study-f: critical correction ratio	draft (working)
11	study-r: reinforcement co scaling	draft (working)
12	study-m: cross domain stability window	draft (working)
13	study-am: form discrimination away from half	draft (working)
14	study-aa: substrate ceiling on correction	draft (working)
15	study-af: gamma monotonicity	draft (working)

16	study-ag: two derivation divergence	draft (working)
18	arc-align-instrument: human scorer validation	draft (working)
19	study-y: cross family rescoring	draft (working)
20	study-t: acoustic time crystal form and engine	draft (working)
21	paper-origin-scaling: frozen exponent ladder	draft (working)
22	study-i: load bearing integrity	draft (working)
23	paper-foundational: cross domain exponent	draft (working)
24	paper-i: recursive depth exponent	draft (working)
25	paper-iii: alignment exponent reassessment	draft (working)
26	paper-vi: complexity scaling replication	draft (working)
27	paper-vi: honey survival analysis	draft (working)
28	paper-xii: public benchmark rescoring	draft (working)
29	study-a: alignment capability scaling	draft (working)
30	study-ab: self acceleration exponent	draft (working)
31	study-aj: balance exponent in genomes	design skeleton
32	study-ak: balance exponent in science	design skeleton
33	study-al: cross substrate margin graduation	design skeleton
34	study-e: recursion form scaling	draft (working)
36	study-j: repurposing resistance	draft (working)
37	study-l: allocation governor	draft (working)
38	study-o: organism and venation scaling	draft (working)
39	study-s5: time crystal exponent	draft (final and ready)
40	paper-v: purpose kernel comparison	draft (working)
41	study-d: embedded vs external correction	draft (working)
42	study-eden: gate position and organism	draft (working)
43	study-g: genesis priority and identity persistence	draft (working)
44	study-n: self maintained correction	draft (working)
45	study-u: genesis governor unification	draft (working)
46	arc-align-instrument: blinding necessity meta analysis	draft (working)
48	arc-align-instrument: depth control equivalence	draft (working)
49	arc-align-instrument: misalignment index validation	draft (final and ready)
50	paper-hrih: registered prediction ledger	draft (working)
51	paper-iv-a: response class stability	draft (working)
52	paper-iv-c: sealed holdout contamination	draft (working)
53	paper-iv-d: blinding evaluation	draft (final and ready)

54	paper-ix: status assignment audit	draft (working)
55	paper-vii: four family logarithmic cell test	draft (working)
56	paper-vii: fresh domain extension	draft (working)
58	paper-xi: convergence independence audit	draft (working)
59	study-eden-b: check evaluability taxonomy	draft (working)
60	study-h: arc align incremental validity	draft (working)
61	study-w: self replication programme	draft (working)
62	study-x: specification curve	draft (working)
63	study-z: cross tradition ethics kernel	draft (working)
64	paper-c: pnp construct validation	draft (working)
65	paper-eden-vision: tradition coverage	draft (working)
66	paper-ii: multi domain extension	draft (working)
67	paper-ii: token calibrated reanalysis	draft (working)
68	paper-iv-b: saturation shape	draft (working)
69	paper-viii: weight level replication	draft (working)
70	study-legal-a: citation proposition support	draft (working)
73	study-v: alignment drift without embedded constraint	draft (working)
·	paper-vii-c: catalogue expansion 80 domain test	draft (working)
·	study-eclipse-m: family ratio grid	design skeleton
·	study-k2: co correction stability	design skeleton

8.4 The printed predictions and their outcomes

The print edition of 2 January 2026 carries nine numbered predictions, five in Appendix F and four in Appendix A, section A.3, with the falsification criteria of section A.4; Section 3 grades each of them. The outcomes register (<https://www.michaeldariuseastwood.com/research/data/prediction-outcomes.json>) records these and the other dated predictions with a deciding instrument as follows (as of 2026-09-07):

PREDICTION	DATED RECORD	STATUS
Book Appendix F Prediction 2: alignment drift above 15 per cent without embedded constraint and below 5 per cent with it, within 18 months	in print 2 January 2026	pending
Book Appendix A.3: AI development scales quadratically rather than linearly with recursive depth	in print 2 January 2026; framework from 8 December 2024	open, undecided
The 12-domain extension: per-row scaling families predicted before fitting (manifest dated 17 March 2026)	17 March 2026: the twelve-row manifest committed 00:19:19Z (8edaeco), twenty-three minutes before the run	fulfilled

The 50-domain tiered suite: per-domain predicted families before fits	16 March 2026 catalogue	supported, exploratory
Temporal out-of-sample: families fitted on old data predict the functional form of new data (7 domains)	March 2026 design	mixed, recorded
The fourth cell: the logarithmic family the corpus has never fitted, on never-fitted domains	prospective design, drafted and awaiting human submission	pending
Acoustic time crystal: super-linear scaling of temporal stability with oscillation cycle count in the transient growth regime (alpha above 1)	10 February 2026, 18:29, self-copied email to the subject lab, the kill staked in the recipient's favour	pending
Quantum error correction scaling (the Willow timing)	8 December 2024 manuscript; the Willow team's preprint public since 24 August 2024	convergent timing only, never a prediction
The early headline exponent: alpha approximately 2.24 (sequential scaling), since retracted	early programme measurement	failed and retracted
Paper III's dated predictions (13 February 2026: F4, F7, F10 to F12), including the leaf-venation $d/(d+1)$ exponent	13 February 2026, Paper III Priority Record section	open, undecided

8.5 Predictions fixed inside designs

Five designs fixed their predictions inside dated files before their runs; the machine-readable register (<https://www.michaeldariuseastwood.com/research/data/design-embedded-predictions-register.json>) lists their artefacts and date bases:

- **The March 2026 preregistration folder (Paper VII extension).** Twelve rows, each with operator_class, predicted_family, predicted_model and target_source fixed before any fit; a written protocol; the author's own checksums; and the OSF component registration TEXT already written. Dated by: commit 8edaeco, 17 March 2026 00:19:19Z, twenty-three minutes before the fits ran at 00:42:41Z; the same manifest was deposited to the public OSF component at 02:06:55Z that night.
- **The 50-domain tiered suite manifest.** Fifty rows, each carrying a predicted family before any fit: 36 power law, 6 exponential, 8 bounded, computed from the file. Dated by: commit 5435e77, 16 March 2026 23:54:53Z, which carries the fifty-row manifest and its results together; the generating rule is earlier, commit 23ce10a at 23:33:19Z (Paper VII v1); the manifest was deposited to the public OSF component at 02:06:42Z on 17 March.
- **The 80-domain expansion catalogue.** Eighty further rows with predicted families (31 power law, 18 exponential, 31 bounded), operator classes, sources and written justifications. Dated by: file contents and repository history; SHA-256 pinned in the staged 80-domain registration.
- **Paper II as the live running record.** The paper was written WHILE the experiments ran: first result 21 January 2026, dated version revisions through v13 by March, each version a timestamped snapshot of what was believed and measured at that moment,

including the version that recorded the 2.24 estimate and the later versions that retracted it. Dated by: the paper's own dated version-history rows.

- **The temporal out-of-sample design.** The design itself is a prediction procedure: fit families on OLD data, predict the functional form of NEW data before examining it. Dated by: run era from the results file.

8.6 The standing wagers

The open wagers, each with its date and several with deadlines that can simply fail, are listed in Section 6.

How to cite this paper

DOI: 10.17605/OSF.IO/C3NTF

APA (AUTHOR-DATE, 7TH EDITION)

Eastwood, M. D. (2026). Register · The ARC Theory · The Dated Prediction Register: Forward Predictions as One Verifiable Chain.
<https://doi.org/10.17605/OSF.IO/C3NTF>.
<https://www.michaeldariuseastwood.com/research/papers/dated-prediction-register.html>

BIBTEX

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  title = {Register · The ARC Theory · The Dated Prediction Register:  
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{https://www.michaeldariuseastwood.com/research/papers/dated-prediction-register.html}  
}
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INLINE

Eastwood, 2026

See </how-to-cite/> (<https://www.michaeldariuseastwood.com/how-to-cite/>) for the canonical citation manual across all artefacts, or </citations.json> (<https://www.michaeldariuseastwood.com/citations.json>) for the machine-readable index.

This paper and Laws IV and V, 3 October 2026. The theory has been stated with five laws since 24 September 2026. Law IV, the ARC Persistence Law, is carried by proposition P15, and Law V, the ARC Embedding Law, by propositions P12 and P21. The three propositions are registered; neither law is registered as a law, and neither has been tested. Their mathematics is set out in **Laws IV and V: the mathematics, version 1.0** (<https://www.michaeldariuseastwood.com/research/reports/laws-iv-v-mathematics-v1-o.html>) (3 October 2026). This register reports no measurement of its own, and it bears on the two laws only as a dated record. Of the nine numbered predictions it records as in print on 2 January 2026, three concern embedded constraint or the persistence of values: Prediction 2 of Appendix F, which section 3 calls preregistration-grade in substance; Prediction 4 of Appendix F, which it calls near-grade; and the fourth prediction of section A.3, which it calls a controlled design in the ordinary sense. The register tests none of the three, and **the theory registration** (<https://doi.org/10.17605/OSF.IO/P8CKQ>) registers none of them as a proposition. In its words, Prediction 2, alignment drift with and without hardware-level constraint, is “the subject of its own drafted unit, with the hardware claim and the eighteen-month window declared untested inside it”. Prediction 4, value stability under adversarial conditions, “is adjacent to P21 and is not the same claim”. The fourth prediction of section A.3 varies the sequencing of value-relevant examples and measures the persistence of the early values against the later ones; **the theory registration** (<https://doi.org/10.17605/OSF.IO/P8CKQ>) calls it the print’s “most precise proposal and the one it never built”, and it too is the subject of a drafted unit. Of the four falsifiers of section A.4, the same registration says of one: “That early-embedded values hold no persistent advantage over later modification is bounded from one side by P15, which measures whether installed gains decay under subsequent training, and is not otherwise decided, because the design the print itself proposes for it has never been built.” In section 6, the five predictions of Appendix F carry the print’s own numbers, prediction 1 to prediction 5; they are not the registered propositions P1 to P5, which Section 8 lists. Nothing in this register is a deciding test of P15, P12 or P21. Naming the two laws changes nothing printed here: the predictions, their dates and their grades stand as they were.

Note (3 October 2026). Nine passages above are changed where they stand, and this note records what each printed before. Section 1 said that the submission click remained outstanding, and the last row of the chain gave the eight draft registrations as awaiting the human click, with the status awaiting human submission. That was the state when this paper was first published on 25 August 2026; both passages now give the state as read on 3 October 2026. Section 3 said that the measurement went the other way, and it gave one figure of the estimator row, the regression exponent, as the blinded cross-architecture estimate; it now gives the row whole, as **Paper II** (<https://doi.org/10.17605/OSF.IO/8FJMA>) reports it, and says that the fit is one model’s and inconclusive. Section 3 also said that Google Willow carries both dates and printed one of them; it now prints both. The first rule of section 1 reserved the word preregistered for an accepted registry submission, and the abstract said that the paper never uses that word for anything except an accepted registry submission; the rule as amended keeps the word registered for an accepted registry submission and calls a design fixed by a dated timestamp before its run preregistered by timestamp, and the abstract follows it. The abstract said without exception that every core quantitative claim has at least one dated artefact fixing it before the measurement that tested it; it now excepts **Paper VII** (<https://doi.org/10.17605/OSF.IO/X6WA7>)’s fifty-domain cohort. Section 3 described the study-v draft registration without the limit that **the theory registration** (<https://doi.org/10.17605/OSF.IO/P8CKQ>) states for it; it now quotes that limit. The chain’s row for 17 March 2026 gave its status as demoted by the author and no more; it now adds the reversal of 7 September 2026, which section 4 sets out.

Working Paper v2.1, 5 October 2026. Version history: v2.1 (5 October 2026): The title now follows the programme’s uniform order (paper, its law or framework, its own title), *Register · The ARC Theory · The Dated Prediction Register: Forward Predictions as One Verifiable Chain*; until this version it read *The Dated Prediction Register: Forward Predictions as One Verifiable Chain*. Section 1 now gives the state of registration as read on 4 October 2026, when the corrector-class audit (unit o6) was registered and public; until this version it gave the state as read on 3 October 2026, when the unit’s approval period at OSF was still running. The register now names the programme it belongs to by the programme’s current name, ARC/Eden Research Programme; until this version it named a series, ARC/Eden experiments. No registered law, prediction, kill

condition or dependency changes; no result is added; no date of any claim moves. v2.0 (4 October 2026): Substantive. A census of every dated prediction in the programme is added as Section 8: the theory registration's twenty-two propositions, quoted with their registers, laws and kill conditions; the registered corrector-class audit (unit 06) with its three registered predictions; the sixty-seven draft designs by unit and stage, with the two held designs and the one withdrawn counted; the outcomes of the printed and other dated predictions; and the predictions fixed inside designs. The chain records unit 06 as registered and public, and the print's five Appendix F predictions are labelled prediction 1 to prediction 5 so that they cannot be read as the registered P1 to P5. The head and social titles now match the citation title, and the citation block gives the DOI, 10.17605/OSF.IO/C3NTF. No registered law, prediction, kill condition or dependency changes; no result is added; no date of any claim moves. v1.3 (3 October 2026): The first vocabulary rule of section 1 is amended: registered is kept for a registry submission that has been accepted, and a design fixed by a dated timestamp before its run is called preregistered by timestamp; the abstract's sentence on the word follows the rule. Section 1 no longer gives the submission click as outstanding, and with the last row of the chain it gives the state of registration as read on 3 October 2026. The abstract excepts **Paper VII** (<https://doi.org/10.17605/OSF.IO/X6WA7>)'s fifty-domain cohort from its claim that every core quantitative claim has a dated artefact fixing it before its measurement. Section 3 gives the estimator row from **Paper II** (<https://doi.org/10.17605/OSF.IO/8FJMA>) whole, as one model's fit and inconclusive; it gives both dates for Google Willow; and it quotes **the theory registration** (<https://doi.org/10.17605/OSF.IO/P8CKQ>)'s limit on the study-v draft. The chain's row for 17 March 2026 adds the reversal of 7 September 2026. The dated note above records what each of those passages printed before, and the note before it places this paper against Laws IV and V. The works the paper cites are linked to their sources. Until this version the paper gave its version as 1.1; it now gives 1.3, because the correction of version 1.2 was published without the paper's stated version being changed. No registered law, prediction, kill condition or dependency changes; no result is added; no date of any claim moves. v1.2 (23 September 2026): The v1.0 history line now names the two verification lanes as the programme's own internal lanes. No claim, date, result or status has changed. v1.1: The provenance note at the end has been corrected for accuracy. No claim, date, result or status has changed. v1.0 first publication, assembled from the prediction register page and its machine register after two-lane verification of every link by the programme's own internal lanes. Corrections to this paper will be dated, listed here, and never silent.

Michael Darius Eastwood conceived and directs this research programme and is the author of this work. Across the programme, he has used more than six AI systems in parallel, under his own instructions, to stress-test his arguments, identify possible errors, and assist in preparing draft text from his own outlines. He determines what is adopted, revised or rejected and takes responsibility for the published content. These systems are tools, not authors.